Department of
Computer Science and Engineering

Lab No. 01

Title: Project Assignment, Proposal Preparation and Planning

Software Engineering Lab

CSE 314

Green University of Bangladesh

1 Objective(s)

- To prepare a project proposal to develop a system including objectives, problem domain, motivations. related works and methodology.

2 Problem analysis and motivations

Problem analysis for example of "ATM Management System" which is an electronic telecommunication system. This system involves the customers of financial institutions like banks, to carry out financial transactions, such as cash withdrawal, Balance Enquiry without any requirement of a bank clerk.

For an ATM Management System, a customer is a person who inserts or swipes plastic card with a magnetic stripe or a plastic smart card with a chip into an ATM containing a different card number. Post inserting the card the client is also asked to give his authentication by entering a special identification number (PIN) which must match with the PIN stored in the chip on the card (if the card has it equipped) or in the financial institution's database who has issued the card.

As huge data is to be maintained, so it is necessary to design and develop a system which will be efficient, error-free, automated and easily maintainable.

The software AMS is to be developed for Automated Teller Machine (ATM). An Automated Teller Machine is computerized system which will provide a secure platform for customers of banks to perform financial transactions in public places such as roads, malls, offices etc.; without any human intervention. This system will provide a user-friendly interface to access financial facilities provided by banks.

3 Literature Review

There are no hard international or government-compiled numbers totaling the complete number of ATMs in use worldwide. Estimates developed by ATMIA place the number of ATMs currently in use at 3 million units, or approximately 1 ATM per 3,000 people in the world [1,2].

To simplify the analysis of ATM usage around the world, financial institutions generally divide the world into seven regions, due to the penetration rates, usage statistics, and features deployed. Four regions (USA, Canada, Europe, and Japan) have high numbers of ATMs per million people [3,4]. Despite the large number of ATMs, there is additional demand for machines in the Asia/Pacific area as well as in Latin America [5,6]. Macau may have the highest density of ATMs at 254 ATMs per 100,000 adults [7]. ATMs have yet to reach high numbers in the Near East and Africa [8].

4 Methodology

Though this system involves a lot of user involvement, new system development must guarantees a system of high quality. It allows users and operators of the new management system to see how the system may look, feel and decide if there are new features to be added to it. By implementing a prototype, the users can see the inputs and outputs from the new system and can decide on what format they want, which can be changed until the user requirements are met, before the final system is built. It is a way of reducing risk, since any error noticed during development can be corrected immediately because the cost of correcting these errors at a later stage can be quite expensive. Methodology is a collection of methods, practices, processes, techniques, procedures, and rules. We can use here different project development methodologies here like agile method, waterfall method, Extreme Programming etc.

5 Feasibility Study

A bunch of feasibility study should be taken to compare the new system going to be build and the existing system. We have to study economical feasibility, Technical feasibility, operational feasibility etc.

5.1 Technical Feasibility

- Calculate System accuracy, Calculate Growth potential rate
- Calculate Response time & Analyze User Friendliness

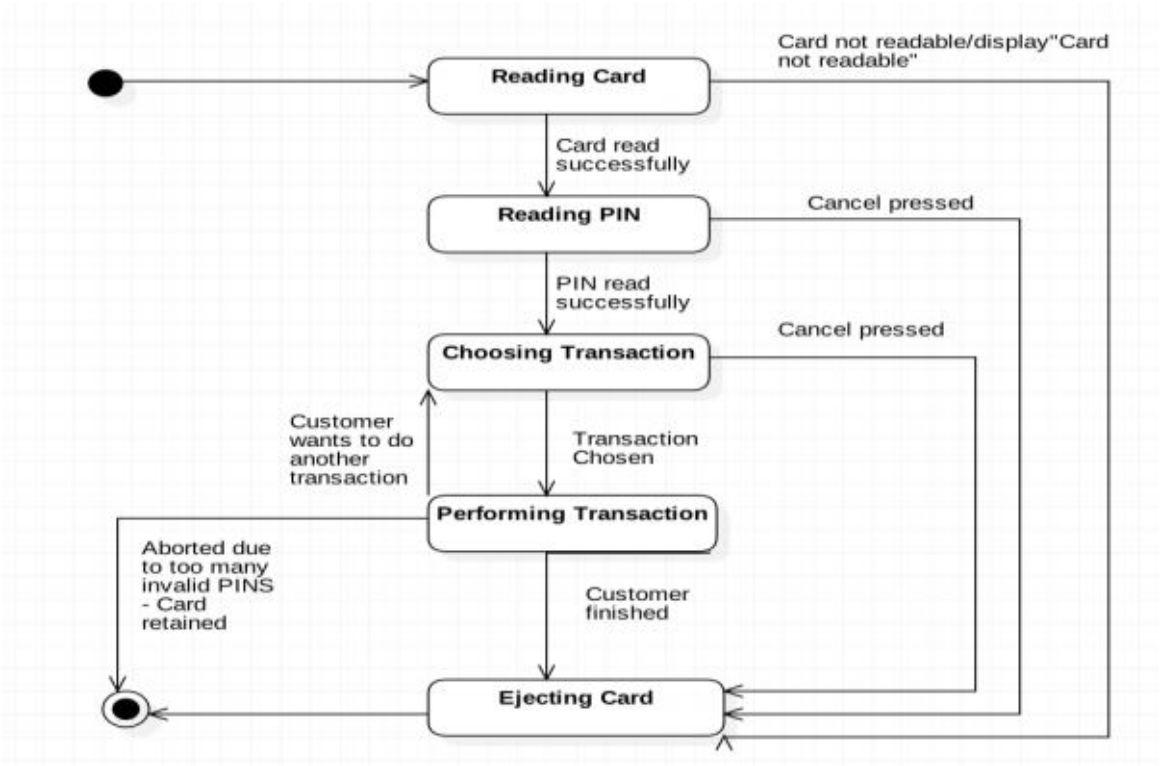

Figure 1: Block Diagram of ATM System

5.2 Operational Feasibility

- Calculate Operation time
- Analyze Reliability, Accuracy & Retrieval Rate

5.3 Economical Feasibility

- Check System performance, Check System operation
- Emphasize User training, Calculate Budget & Check Human resources

6 Main Phases

- Project proposal and planning.
- Requirement specification of a project.
- SDLC selection for a specific project.
- Developing data flow diagram (DFD) model of a project.
- Develop UML use case diagram for the given project.
- Develop UML sequence and communication diagram for the given project.
- Develop UML class diagram for the given project.
- Software testing.

7 Detailed Working Plan

See the following detailed working plans:

SL	Task	Required Week	Responsible person	Phase
1	Requirement Specification and Data Collection	1	Project Manager and team members	Research and Planning
2	Requirement Finalization	1	Project Manager and team members	Analysis
3	System Design and Modeling	1	Project Manager and team members	Design
4	System Modeling and Finalization	1	Project Manager and team members	Design
5	System Development (Coding)	1	Project Manager and team members	Implementation
6	Testing and Feedback sharing	1	Project Manager and team members	Testing
7	Beta Version Delivery for Feedback	1	Project Manager and team members	Testing
8	Feedback sharing and Requirement Change	1	Project Manager and team members	Testing
9	Delivery preparation with documentation	1	Project Manager and team members	Deployment

8 Gantt Chart of Project Development Timing

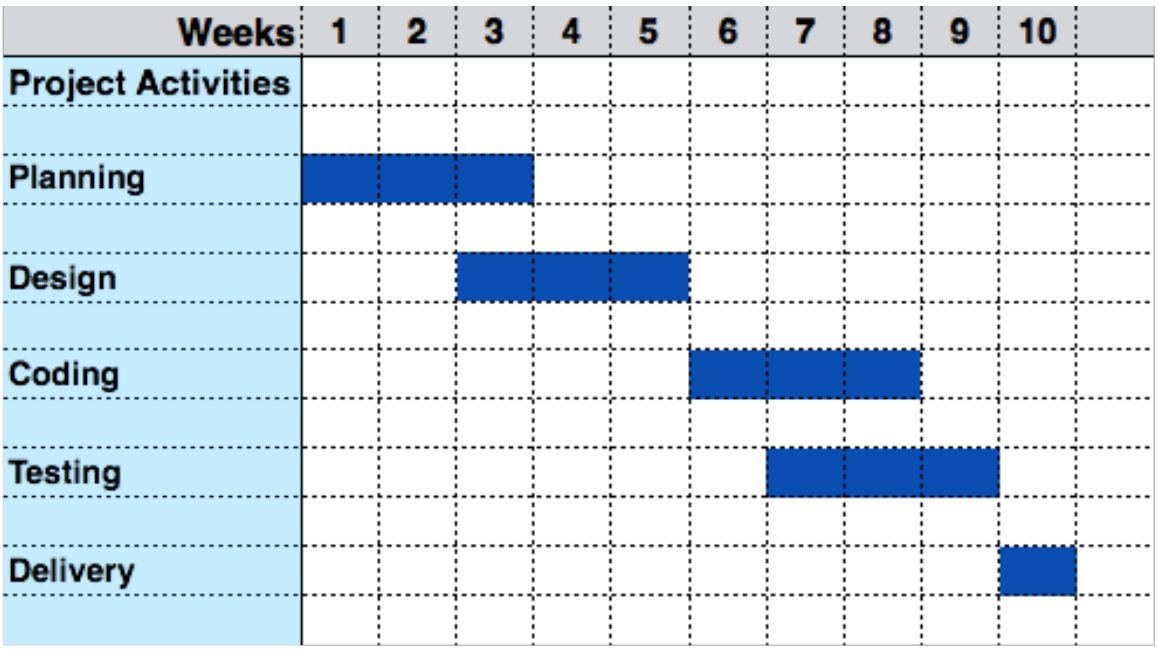

Figure 2: Project Development Timing

9 Budget details of an ATM banking system

See the following Budget details (Table 1)

Table 1: Budget Details

SL	Criteria	Cost specification	Existing system(tk)	New system(tk)
1	Office Cost	Team meeting	25,000	20,000
		Project meeting	25,000	20,000
		First aid	1000	500
2	Website Cost	Website maintenance	2,000	1,500
3	Office equipment Cost	Computer	10,00,000	9,50,000
		Laptop	2,00,000	1,50,000
		ATM machine	3,00,000	2,50,000
		CC Camera	1,00,000	1,00,000
4	Salary Cost	Team Leader	2,00,000	2,50,000
		System Designer	80,000	70,000
		Software Engineer	1,00,000	1,00,000
		Animator	60,000	60,000
		Developer	50,000	50,000
		Officer	40,000	38,000
		Sentry	10,000	9,000
		Total cost	21,93,000	20,69,000

10 Discussion & Conclusion

Based on the focused objective(s), to understand about project proposal planning, the additional lab exercise made us more confident towards the fulfilment of the objectives(s).

11 Lab Task (Please implement yourself and show the output to the instructor)

1. Prepare a detailed budget of a library management system.
2. Prepare a timelines of library management system using gantt chart.
3. Show the required information of library management project phases.

12 Lab Exercise (Submit as a report)

- Think about a new system with your partners and submit a report touching the above criteria.

13 Policy

Copying from internet, classmate, seniors, or from any other source is strongly prohibited. 100% marks will be *deducted* if any such copying is detected.

References

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